Ride-Height and Pitch Dynamics of a Fully Submerged Hydrofoil Boat: Nonlinear Model, Trim and LTI State-Space Form

ELEC3304/9304 Control — Practical Week 9 / Assessment 2
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October 2026

1 System and control objective

The plant is a small single-seat hydrofoil boat of the solar-boat class, with parameters taken from the Técnico Solar Boat SR02 [1]. Above its take-off speed the hull is carried clear of the water by three fully submerged foils: two at the front (port and starboard, treated here as one lifting surface) and one at the rear on the propulsion strut.

A fully submerged foil, unlike a surface-piercing one, has almost no natural tendency to return to a given depth: its lift is nearly independent of how deep it runs. The boat therefore cannot hold a ride height by itself and needs active control [1]. The control objective is to regulate the ride height at a set value and keep the hull level, in the presence of waves, using only small changes of the foil incidence angles.

We consider the longitudinal (vertical-plane) motion only: heave and pitch.

2 Signals

Role	Symbol	Meaning	Unit
State	h	heave: height of the centre of mass (CM) above its trim height, positive up	m
State	$v = \dot{h}$	vertical velocity of the CM, positive up	m/s
State	θ	pitch angle, positive bow-up	rad
State	$q = \dot{\theta}$	pitch rate	rad/s
Input	δ_f	incidence of the front foils relative to the hull	rad
Input	δ_r	incidence of the rear foil relative to the hull	rad
Output	y_1	ride height measured by an ultrasonic sensor at the bow	m
Output	y_2	pitch angle measured by an attitude reference unit (AHRS)	rad
Output	y_3	pitch rate measured by a rate gyro	rad/s
Disturbance	w_f, w_r	vertical water velocity (wave orbital motion) at the front / rear foil	m/s
Disturbance	ζ	wave elevation under the height sensor	m
Noise	n_1, n_2, n_3	sensor noise	m, rad, rad/s

There are $n_x = 4$ states, $n_u = 2$ actuators and $n_y = 3$ sensors, so the system has fewer measurements than states and the unmeasured states must be estimated. The sensors are the

ones fitted to the real boat [1]: an ultrasonic height sensor at the bow and an attitude and heading reference system (AHRS), which provides both the pitch angle and the pitch rate. Heave and heave rate are not measured; the choice of sensors is justified in Section 8.3.

3 Modelling assumptions

A1. The boat is foil-borne; the hull does not touch the water.

A2. Motion is confined to the vertical plane. Roll, sway and yaw are zero.

A3. The forward speed U is constant. On the real boat a separate speed loop acts on the throttle [1]; variations of U are treated as model uncertainty in Section 10.

A4. Foil drag and propeller thrust are neglected in the heave and pitch equations. At trim they cancel in the surge direction and their net pitching moment is about 20 N m, roughly 1 % of the 1600 N m moment of each foil group about the CM. The change of drag with angle of attack is a few percent of the change of lift.

A5. Foil lift is quasi-steady, with a lift coefficient that is affine in the angle of attack, $C_L(\alpha) = C_{L0} + C_{L\alpha}\alpha$, valid below stall ($|\alpha| \lesssim 12°$, assumed).

A6. Hydrodynamic added mass of the foils is neglected. Estimated as $\frac{\pi}{4}\rho c^2 b$ per foil it totals about 8 kg, i.e. 4 % of the boat mass and, at the foil positions, about 6 % of I_y.

A7. The front foils move together (common mode). The rear foil is assumed to be actuated; on SR02 it was fixed and its actuation was listed as future work [1].

A8. Actuator dynamics are neglected in the design model; position and rate limits are kept as constraints (Section 9). In simulation each foil actuator is modelled as a position limit, a first-order servo lag with time constant τ_a, and a rate limit.

4 Nonlinear model

4.1 Kinematics

Each foil group $i \in \{f, r\}$ is located, in hull axes, a distance x_i ahead of the CM and z_i below it, with $x_f = l_f > 0$ and $x_r = -l_r < 0$. The foils sit well below the CM, so z_i cannot be ignored. When the hull is pitched by θ the foil position relative to the CM, in horizontal/vertical axes, is

$$X_i = x_i \cos\theta + z_i \sin\theta, \qquad Z_i = x_i \sin\theta - z_i \cos\theta. \tag{1}$$

Differentiating, and subtracting the vertical water velocity w_i, gives the velocity of each foil relative to the water:

$$u_i = U + q\,(z_i \cos\theta - x_i \sin\theta), \qquad v_i = v + q\,X_i - w_i. \tag{2}$$

The inflow angle is $\gamma_i = \arctan(v_i/u_i)$. A foil that moves upward sees the flow coming from above, which reduces its angle of attack:

$$\alpha_i = \delta_i + \theta - \gamma_i. \tag{3}$$

The term $-\gamma_i$ is what provides heave and pitch damping. It is absent from the model of [1], which uses $\alpha = \delta + \theta$; we follow [2, 3] and keep it.

With d_0 the submergence of the foils at trim, the instantaneous submergence is

$$d_i = d_0 - h - x_i \sin\theta - z_i\,(1 - \cos\theta). \tag{4}$$

4.2 Foil lift with free-surface effect

The lift of foil group i with planform area S_i is

$$L_i = \tfrac{1}{2}\,\rho\,(u_i^2 + v_i^2)\,S_i\,K\!\left(\frac{d_i}{c}\right)\big(C_{L0} + C_{L\alpha}\,\alpha_i\big), \tag{5}$$

where c is the foil chord and K accounts for the loss of lift near the free surface. The depth Froude number of this boat is $U_0/\sqrt{g d_0} \approx 4.1$, so we use the infinite-Froude-number ("biplane") result of linear theory, derived in Appendix A (see also [4], ch. 6),

$$K(\xi) = \frac{1 + 16\,\xi^2}{2 + 16\,\xi^2}, \qquad K'(\xi) = \frac{32\,\xi}{\big(2 + 16\,\xi^2\big)^2}, \qquad \xi = \frac{d}{c}. \tag{6}$$

$K \to 1$ for a deeply submerged foil and $K \to \tfrac{1}{2}$ as the foil reaches the surface; in simulation K is set to zero once a foil leaves the water. Because $K' > 0$, a boat that rises loses lift, which gives a (weak) restoring force in heave.

4.3 Equations of motion

Lift acts perpendicular to the relative flow, so foil i applies the force $(-L_i \sin\gamma_i,\ L_i \cos\gamma_i)$ at the point (X_i, Z_i). Newton's second law in the vertical direction and about the pitch axis gives

$$m\,\dot v = \sum_{i\in\{f,r\}} L_i \cos\gamma_i \;-\; mg, \tag{7}$$

$$I_y\,\dot q = \sum_{i\in\{f,r\}} L_i\,\big(X_i \cos\gamma_i + Z_i \sin\gamma_i\big), \tag{8}$$

together with $\dot h = v$ and $\dot\theta = q$. With $\boldsymbol{x} = [\,h \;\; v \;\; \theta \;\; q\,]^\top$, $\boldsymbol{u} = [\,\delta_f \;\; \delta_r\,]^\top$ and $\boldsymbol{w} = [\,w_f \;\; w_r\,]^\top$ this is a nonlinear system $\dot{\boldsymbol{x}} = \boldsymbol{f}(\boldsymbol{x}, \boldsymbol{u}, \boldsymbol{w})$. The nonlinearities are the trigonometric terms, the dependence of dynamic pressure on the foil velocity, the free-surface factor K, and (in simulation) stall and actuator saturation.

4.4 Measurement

The ultrasonic sensor is mounted a distance l_s ahead of the CM and measures the height of the bow above the water; the AHRS measures the pitch angle and the pitch rate. Relative to the trim readings,

$$y_1 = h + l_s \sin\theta - \zeta(l_s, t) + n_1, \qquad y_2 = \theta + n_2, \qquad y_3 = q + n_3. \tag{9}$$

The height sensor sees a mixture of heave and pitch, not either state alone, and the wave surface ζ under the bow enters its reading directly. Unlike n_1, ζ is not white: it is concentrated near the wave encounter frequency.

5 Parameters

* Not reported in [1]. The reported slope 0.090/deg, with the finite-span correction $C_L = C_l\,\mathrm{AR}/(\mathrm{AR}+2)$ used there and a section slope of 2π rad^{-1}, implies an aspect ratio AR ≈ 9, hence $c = \sqrt{S/\mathrm{AR}} \approx 0.073$ m for one front foil. The same chord is assumed for the rear foil.
† Not reported. Chosen so that the foils run about four chords deep with the hull 0.3–0.4 m above the water. Its effect is examined in Section 10.

Symbol	Description	Value	Source
m	mass of boat and pilot	$215\,\mathrm{kg}$	[1]
I_y	pitch moment of inertia	$503\,\mathrm{kg\,m^2}$	[1]
S_f	front foil area ($2 \times 0.048\,\mathrm{m^2}$)	$0.096\,\mathrm{m^2}$	[1]
S_r	rear foil area	$0.046\,\mathrm{m^2}$	[1]
l_f	CM to front foils	$1.10\,\mathrm{m}$	[1]
l_r	CM to rear foil	$2.45\,\mathrm{m}$	[1]
z_f	front foils below CM	$0.75\,\mathrm{m}$	[1]
z_r	rear foil below CM	$0.96\,\mathrm{m}$	[1]
l_s	CM to ultrasonic sensor	$3.0\,\mathrm{m}$	[1]
C_{L0}	lift coefficient at zero incidence (NACA 63-412)	0.288	[1]
$C_{L\alpha}$	lift-curve slope, 0.090 per degree	$5.157\,\mathrm{rad^{-1}}$	[1]
U_0	cruise speed	$7\,\mathrm{m/s}$	[1]
ρ	sea-water density	$1025\,\mathrm{kg/m^3}$	[1]
g	gravitational acceleration	$9.81\,\mathrm{m/s^2}$	—
c	foil chord	$0.073\,\mathrm{m}$	estimated*
d_0	foil submergence at trim	$0.30\,\mathrm{m}$	assumed†
α_{stall}	angle of attack beyond which lift stops rising	12°	assumed
τ_a	servo time constant (simulation only)	$0.02\,\mathrm{s}$	assumed
σ_y	height sensor noise, std	$1\,\mathrm{cm}$	assumed
σ_θ	AHRS pitch-angle noise, std	0.3°	assumed
σ_q	rate gyro noise, std	$0.2\,^\circ/\mathrm{s}$	assumed

6 Equilibrium (trim)

We look for a steady level flight condition in calm water: $\boldsymbol{x}_0 = \mathbf{0}$, $\boldsymbol{w} = \mathbf{0}$, $U = U_0$. Setting $\dot{v} = \dot{q} = 0$ in (7)–(8) gives a force and a moment balance,

$$L_{f0} + L_{r0} = mg, \qquad l_f L_{f0} = l_r L_{r0},$$

so the weight is shared in inverse proportion to the lever arms:

$$L_{f0} = \frac{l_r}{l_f + l_r}\, mg = 1455.6\,\mathrm{N}, \qquad L_{r0} = \frac{l_f}{l_f + l_r}\, mg = 653.5\,\mathrm{N}. \tag{10}$$

Inverting (5) with $u_i = U_0$, $v_i = 0$ gives the trim incidences in closed form:

$$\delta_{i0} = \frac{1}{C_{L\alpha}} \left(\frac{L_{i0}}{\frac{1}{2}\rho U_0^2 S_i K_0} - C_{L0} \right), \qquad K_0 = K(d_0/c) = 0.9963, \tag{11}$$

which yields $\delta_{f0} = 3.53^\circ$ and $\delta_{r0} = 3.11^\circ$ at $U_0 = 7\,\mathrm{m/s}$.

Validation. Table 1 compares (11) with the trim angles that [1] obtained numerically from a model that includes drag and thrust. The agreement supports assumption A4.

7 Linearisation

Define deviations from trim, $\tilde{\boldsymbol{u}} = \boldsymbol{u} - \boldsymbol{u}_0$ (the states and disturbances are already zero at trim). To first order,

$$X_i \approx x_i + z_i\theta, \quad Z_i \approx -z_i, \quad u_i \approx U_0 + z_i q, \quad \gamma_i \approx \frac{v + x_i q - w_i}{U_0}, \quad d_i \approx d_0 - h - x_i\theta.$$

Table 1: Trim incidences versus speed: this model and [1].

U (m/s)	δ_{f0} (deg)	δ_{f0} [1]	δ_{r0} (deg)	δ_{r0} [1]
6	5.97	5.9	5.39	5.5
7	3.53	3.5	3.11	3.2
8	1.96	1.9	1.63	1.7
9	0.87	0.8	0.62	0.7
10	0.10	0.1	−0.11	0.0

The lift perturbation of foil group i is therefore

$$\tilde{L}_i = k_i\Big(\tilde{\delta}_i + \theta - \frac{v + x_i q - w_i}{U_0}\Big) - \kappa_i\,(h + x_i\theta) + \sigma_i\,q, \tag{12}$$

with three coefficients per foil group,

$$k_i = \left.\frac{\partial L_i}{\partial \alpha_i}\right|_0 = \tfrac{1}{2}\rho U_0^2 S_i K_0 C_{L\alpha}, \qquad \kappa_i = \left.\frac{\partial L_i}{\partial d_i}\right|_0 = \frac{L_{i0}}{c}\,\frac{K'(d_0/c)}{K_0}, \qquad \sigma_i = \left.\frac{\partial L_i}{\partial u_i}\right|_0 z_i = \frac{2L_{i0}z_i}{U_0}. \tag{13}$$

Here k_i is the lift per unit angle of attack, κ_i the lift gained per metre of extra submergence, and σ_i the lift gained because a pitch rate moves a foil below the CM forward through the water. Numerically $k_f = 12\,386\,\mathrm{N/rad}$, $k_r = 5935\,\mathrm{N/rad}$, $\kappa_f = 35.5\,\mathrm{N/m}$, $\kappa_r = 15.9\,\mathrm{N/m}$, $\sigma_f = 312\,\mathrm{N\,s/rad}$ and $\sigma_r = 179\,\mathrm{N\,s/rad}$.

Linearising (7) and (8), and using the trim balance $\sum_i L_{i0}x_i = 0$, gives

$$m\,\dot{v} = \sum_i \tilde{L}_i, \tag{14}$$

$$I_y\,\dot{q} = \sum_i x_i\tilde{L}_i \;+\; W_z\,\theta \;-\; \sum_i L_{i0}z_i\,\frac{v + x_i q - w_i}{U_0}, \qquad W_z = \sum_i L_{i0}z_i = 1719\,\mathrm{N\,m}. \tag{15}$$

The last two terms of (15) come from the vertical offset of the foils. $W_z\theta$ is an inverted-pendulum moment: the weight is carried at points below the CM, so pitching the hull moves the lift forward. The final term is the moment of the backward tilt of the lift vector when a foil rises. The two cancel when the boat climbs along its own axis ($v = U_0\theta$), as they must.

Collecting terms gives the LTI model

$$\dot{\boldsymbol{x}} = A\boldsymbol{x} + B\tilde{\boldsymbol{u}} + E\boldsymbol{w}, \qquad y = C\boldsymbol{x} + D\tilde{\boldsymbol{u}} + n, \tag{16}$$

$$A = \begin{bmatrix} 0 & 1 & 0 & 0 \\ -\dfrac{\sum\kappa_i}{m} & -\dfrac{\sum k_i}{mU_0} & \dfrac{\sum(k_i - \kappa_i x_i)}{m} & \dfrac{\sum(\sigma_i - k_i x_i/U_0)}{m} \\ 0 & 0 & 0 & 1 \\ -\dfrac{\sum\kappa_i x_i}{I_y} & -\dfrac{\sum k_i x_i + W_z}{I_y U_0} & \dfrac{\sum(k_i x_i - \kappa_i x_i^2) + W_z}{I_y} & -\dfrac{\sum(k_i x_i^2 - L_{i0}z_i x_i)}{I_y U_0} \end{bmatrix}, \tag{17}$$

$$B = \begin{bmatrix} 0 & 0 \\ \dfrac{k_f}{m} & \dfrac{k_r}{m} \\ 0 & 0 \\ \dfrac{k_f x_f}{I_y} & \dfrac{k_r x_r}{I_y} \end{bmatrix}, \qquad E = \frac{1}{U_0}\begin{bmatrix} 0 & 0 \\ \dfrac{k_f}{m} & \dfrac{k_r}{m} \\ 0 & 0 \\ \dfrac{k_f x_f + L_{f0}z_f}{I_y} & \dfrac{k_r x_r + L_{r0}z_r}{I_y} \end{bmatrix}, \tag{18}$$

$$C = \begin{bmatrix} 1 & 0 & l_s & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix}, \qquad D = \begin{bmatrix} 0 & 0 \\ 0 & 0 \\ 0 & 0 \end{bmatrix}. \tag{19}$$

All sums run over $i \in \{f, r\}$.

Numerical values at $U_0 = 7\,\mathrm{m/s}$ (angles in radians):

$$A = \begin{bmatrix} 0 & 1 & 0 & 0 \\ -0.2394 & -12.173 & 85.213 & 2.8932 \\ 0 & 0 & 0 & 1 \\ 0 & -0.2281 & 1.3208 & -14.470 \end{bmatrix}, \quad B = \begin{bmatrix} 0 & 0 \\ 57.609 & 27.604 \\ 0 & 0 \\ 27.087 & -28.908 \end{bmatrix}, \quad E = \begin{bmatrix} 0 & 0 \\ 8.2298 & 3.9435 \\ 0 & 0 \\ 4.1796 & -3.9515 \end{bmatrix}, \tag{20}$$

with $l_s = 3\,\mathrm{m}$ in C. The entry A_{41} is exactly zero because $\kappa_i \propto L_{i0}$ and $\sum L_{i0} x_i = 0$. Per degree of foil incidence, the front foils produce $1.01\,\mathrm{m/s^2}$ of heave acceleration and $0.47\,\mathrm{rad/s^2}$ of pitch acceleration; the rear foil produces $0.48\,\mathrm{m/s^2}$ and $-0.50\,\mathrm{rad/s^2}$.

Check. The matrices (20) were compared with a central-difference Jacobian of the full nonlinear model (1)–(8) at the trim point; the largest difference in any entry of A, B or E is below 10^{-8}.

8 Open-loop properties

8.1 Modes

The eigenvalues of A are all real:

$$\lambda_1 = -14.81, \qquad \lambda_2 = -11.80, \qquad \lambda_3 = -0.067, \qquad \lambda_4 = +0.027.$$

- **Fast modes** ($\lambda_{1,2}$, time constants $\approx 0.07\,\mathrm{s}$ and $0.08\,\mathrm{s}$): the angle-of-attack response in pitch and heave. Any vertical or pitch velocity changes the foil angle of attack and is damped quickly.
- **Slow ride-height modes** ($\lambda_{3,4}$): two poles very close to the origin. One is weakly stable; the other is *unstable*, with a doubling time of $\ln 2/0.027 \approx 25\,\mathrm{s}$.

The plant is therefore open-loop unstable. The physical reason is that a fully submerged foil provides almost no ride-height stiffness: the only restoring force is the weak free-surface term $\kappa_f + \kappa_r \approx 51\,\mathrm{N/m}$. If the free-surface effect is removed ($\kappa_i = 0$) the two slow poles collapse to a double pole at the origin, so the height dynamics are close to a double integrator. As an illustration of how weak the stiffness is, a 1 % increase in speed raises the total lift by about 42 N; a height change of order $42/51 \approx 0.8\,\mathrm{m}$ would be needed for κ to absorb it, far more than the $0.3\,\mathrm{m}$ submergence. Feedback is required to hold a ride height.

8.2 Reachability

The reachability matrix $\mathcal{C} = [\,B \;\; AB \;\; A^2B \;\; A^3B\,]$ has rank 4, so the system is reachable. It is also reachable from either actuator alone (rank 4 with only the first or only the second column of B), which matters if one actuator fails or is fixed as on SR02.

8.3 Observability and choice of sensors

The observability matrix

$$\mathcal{O} = \big[\,C^\top \;\; (CA)^\top \;\; (CA^2)^\top \;\; (CA^3)^\top\,\big]^\top$$

has rank 4, so all four states can be reconstructed and an observer can be designed. The rank is also 4 with any one of the three sensors alone, so rank alone does not tell us which sensors are needed. Two further questions decide it: how accurately the states can be estimated with noise, and whether the estimates stay unbiased when the model is wrong.

Accuracy. Table 2 gives the steady-state error of a Kalman filter for three sensor sets, with the noise levels of Section 5 and wave velocities of about 0.08 m/s rms at the foils. Adding the pitch rate reduces the pitch errors by a factor of about three; adding the pitch angle halves the pitch-angle error again. The heave estimates depend mainly on the height sensor and change little.

Bias under model error. A speed or mass different from the design value changes the front and rear lift in the same proportion, so it acts on the boat as an unknown constant vertical force. With only the height and pitch-rate sensors, such a force cannot be told apart from a small pitch offset: the bow height reading is the same either way. In closed-loop simulation with the best two-sensor design we found (*control_design.pdf*, D2), a speed 1 m/s below the design value left the boat 12.6 cm too low and pitched by 2.4° in steady state, while the observer reported both as zero. No observer tuning removed this. Once the pitch angle is measured, the force becomes clearly observable: an observer with one extra state for the force, and integral action on height and pitch in the controller, bring both errors to zero.

Table 2: Estimation error of a Kalman filter (one standard deviation) and steady-state error in closed loop for a 1 m/s speed error, for different sensor sets.

		Estimation error				Steady error, $U = 6$ m/s
Sensors	n_y	h (cm)	v (cm/s)	θ (deg)	q (deg/s)	h, θ
Height only	1	0.92	5.2	0.22	1.48	—
Height + pitch rate	2	0.90	4.3	0.06	0.47	−12.6 cm, 2.4°
All three (chosen)	3	0.90	4.3	0.03	0.47	0, 0

We therefore use all three measurements. This matches the real boat, which carries an AHRS [1], keeps $n_y < n_x$, and still requires an observer for the heave and the heave rate, which are not measured.

9 Input constraints

Constraint	Value	Source
Front foil incidence, absolute	$-3° \le \delta_f \le 7°$	mechanical range [1]
Front foil incidence, deviation from trim	$-6.5° \le \tilde{\delta}_f \le 3.5°$	from $\delta_{f0} = 3.53°$
Rear foil incidence, deviation from trim	$-6.1° \le \tilde{\delta}_r \le 3.9°$	same range assumed
Incidence rate (stepper motor)	$\lvert\dot{\delta}_i\rvert \le 20\,°/\mathrm{s}$	assumed; to be confirmed

The limits matter for the controller design in two ways. First, the limits are *asymmetric* about the operating point: the boat has almost twice as much authority to reduce lift as to increase it. Second, the available authority depends on speed. From Table 1, the trim incidence already reaches about 6° at 6 m/s, leaving only 1° of upward margin, and below about 5.7 m/s level flight is not possible within the 7° limit. A rate limit was identified in [1] as the cause of a loss of height in simulation; its numerical value is not reported there.

10 Model uncertainty

The dependence on speed and on submergence was evaluated with the same formulas:

Source	Range considered	Effect on the model
Forward speed U	6–8 m/s	$B \propto U^2$; fast-mode poles roughly $\propto U$; trim shifts
Mass m (pilot, batteries)	±10 %	trim lift, hence κ_i, σ_i, W_z (rows 2 and 4 of A, and E); $1/m$ in row 2
Pitch inertia I_y	±10 %	row 4 of A and B
Lift slope $C_{L\alpha}$	±10 %	k_f, k_r (B, E and the damping terms of A); trim incidences
Trim submergence d_0	0.15–0.40 m	heave stiffness κ_i; location of the slow poles
Foil added mass	+4 % of m	neglected (A6)
Stall, foil ventilation	large excursions	lift lower than the linear model predicts

Table 3: Open-loop eigenvalues for different speeds (left, $d_0 = 0.30$ m) and trim submergences (right, $U = 7$ m/s).

U (m/s)	fast modes	slow modes	d_0 (m)	fast modes	slow modes
6	−12.82, −10.00	−0.086, +0.039	0.15	−14.54, −11.51	−0.292, −0.020
7	−14.81, −11.80	−0.067, +0.027	0.20	−14.70, −11.69	−0.153, +0.021
8	−16.74, −13.65	−0.050, +0.016	0.30	−14.81, −11.80	−0.067, +0.027
			0.40	−14.84, −11.83	−0.039, +0.022

The fast modes are insensitive to d_0 and scale almost linearly with U. The slow modes stay within 0.3 rad/s of the origin in every case, but the sign of the smaller one changes: it is unstable for most of the range and becomes weakly stable when the foils run very shallow ($d_0 = 0.15$ m), where the free-surface stiffness is larger. Whether the open-loop plant is weakly stable or weakly unstable therefore depends on parameters we had to assume (c and d_0), while the conclusion that the height behaves almost like a double integrator, and needs feedback, does not. The input gain changes by a factor $(8/6)^2 \approx 1.8$ across the speed range, which is the main uncertainty a fixed-gain controller will have to tolerate.

11 Open-loop simulation

The model was simulated in two independent ways, with no controller:

- **MATLAB**: the nonlinear model (1)–(8) is integrated with `ode45`, and the LTI model with `lsim`.
- **Simulink** (`hydrofoil_sim.slx`): actuators (position limit, servo lag, rate limit), nonlinear boat, the LTI model run in parallel, waves, and the three sensors with noise. The controller is switched off for these runs.

All inputs below are deviations from the trim incidences, and every run starts at trim. Table 4 confirms that the two implementations agree.

Table 4: Agreement between the Simulink model and independent MATLAB simulations.

Test	Comparison	Largest heave difference
Front foil step, ideal actuators	Simulink nonlinear vs `ode45`	9.5×10^{-11} m
Regular waves, ideal actuators	Simulink nonlinear vs `ode45`	9.5×10^{-7} m
Regular waves	Simulink linear vs `lsim`	1.4×10^{-5} m

11.1 Response to step, pulse and sine inputs

Figure 1 applies three small inputs. Linear and nonlinear responses are indistinguishable, which confirms the linearisation. The responses also show the open-loop character found in Section 8.1:

- A step of only 0.1° on the front foils makes the pitch angle grow steadily (about 0.17 °/s) and the height grow quadratically, reaching 11 cm three seconds later. The height dynamics behave like a double integrator.
- A short pulse on the rear foil leaves the boat with a constant pitch offset of −0.4° and a steady descent of about 4.5 cm/s. Nothing brings the boat back.
- A sinusoidal input produces an oscillation superimposed on the same drift.

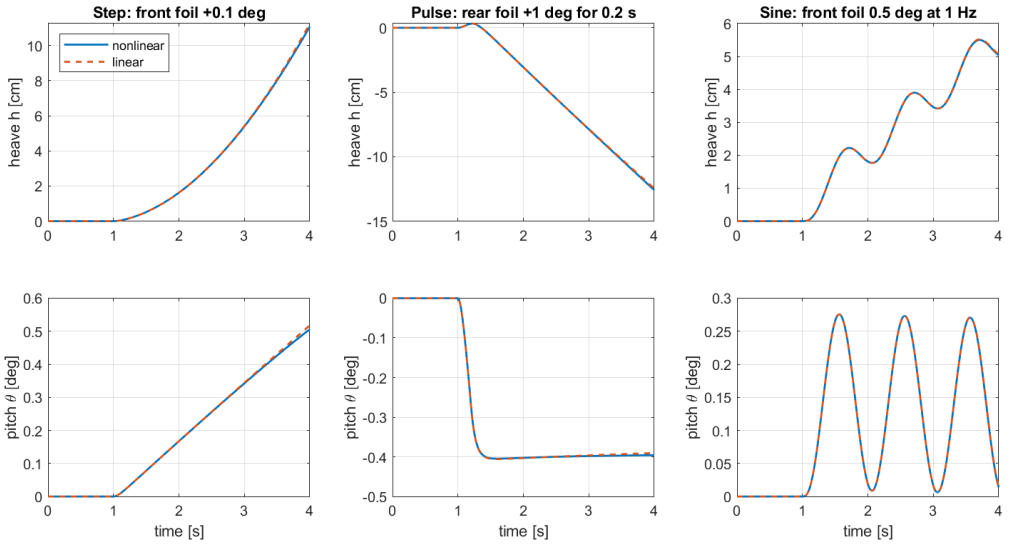


Figure 1: Linear and nonlinear responses to a front-foil step, a rear-foil pulse and a front-foil sine, all applied at $t = 1\,\mathrm{s}$.

11.2 Range of validity of the linear model

Figure 2 repeats the front-foil step with increasing size. The two models agree until the heave reaches about 15 cm. Beyond that the foils approach the surface, the free-surface factor K falls quickly, and the nonlinear model levels off at 24–27 cm while the linear model keeps climbing. The model has no ventilation, so the behaviour this close to the surface is only indicative; in reality the foils would be likely to ventilate and lose more lift. We take the linear model as adequate for $|h| \lesssim 15\,\mathrm{cm}$, and the controller will have to keep the boat in that range.

11.3 Actuator constraints

In Figure 3 the front foils are commanded 5° above trim, beyond the 3.5° that the mechanical limit allows. The applied angle stops at 7° and takes about 0.17 s to get there because of the rate limit. The height response is delayed by about 0.15 s. The upward margin of 3.5° is therefore the binding constraint whenever the controller needs more lift, for example to recover from a drop.

11.4 Model uncertainty

Figure 4 keeps the foils at the nominal trim incidences while the true speed or mass differs from the design values. A speed error of 0.2 m/s (about 3 %) moves the boat by about 20 cm in 8 s;

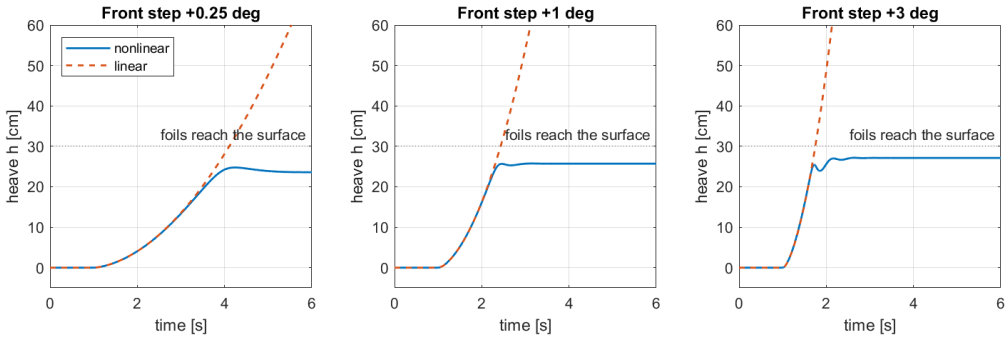


Figure 2: Front-foil steps of 0.25°, 1° and 3°. The dotted line marks the height at which the foils reach the surface.

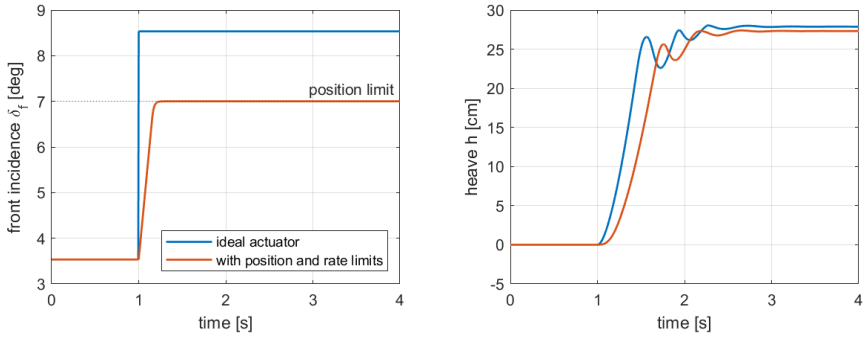


Figure 3: A front-foil command beyond the position limit, with an ideal actuator and with position and rate limits.

a mass error of 10 kg (about 5 %) by about 15–17 cm. A fixed trim setting cannot hold the ride height under realistic parameter errors, so the controller will need integral action.

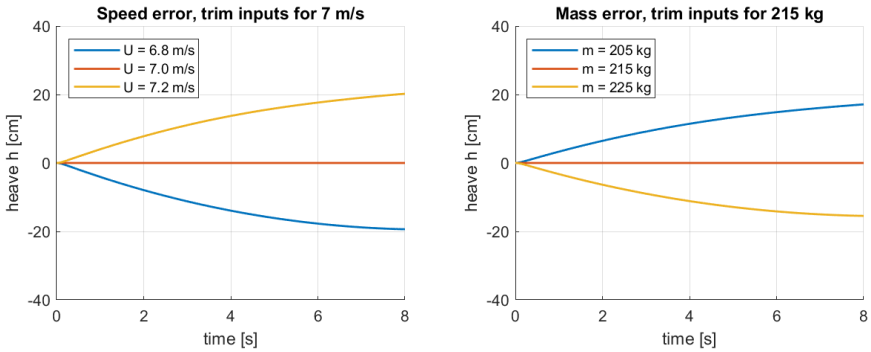


Figure 4: Heave with the inputs fixed at the nominal trim, for speed errors (left) and mass errors (right).

11.5 Waves and measurements

The waves are described by linear (Airy) wave theory in deep water. A component of amplitude a, frequency ω and wave number $k = \omega^2/g$ travelling towards the bow has, in a frame moving with the boat,

$$\zeta(x,t) = a\cos\big(kx + \omega_e t + \phi\big), \qquad w(x,t) = -a\,\omega\, e^{-kd_0}\sin\big(kx + \omega_e t + \phi\big), \qquad \omega_e = \omega + kU, \quad (21)$$

where ζ is the surface elevation, w the vertical water velocity at the foil depth d_0, x the position along the hull and ω_e the encounter frequency. The foils see w at $x = x_f$ and $x = x_r$, and the

height sensor sees ζ at $x = l_s$. A regular wave is a single component; an irregular sea is a sum of components drawn from a wave spectrum.

Figure 5 shows the response to a regular head sea with amplitude 5 cm and wavelength 8 m. At 7 m/s the encounter frequency is 8.27 rad/s (1.32 Hz) and the vertical water velocity at foil depth has an amplitude of 0.11 m/s. The pitch angle oscillates by about $\pm 0.37°$ around a mean of about $-0.18°$, and the heave oscillates while drifting downward by about 20 cm in 10 s; the linear and nonlinear models start to separate once the drift exceeds about 15 cm, as in Figure 2.

The right panel shows the bow height that the ultrasonic sensor would report. The wave surface passing under the bow and the sensor noise are of the same size as the boat motion itself. The measurement cannot be fed back directly; it has to be filtered by an observer.

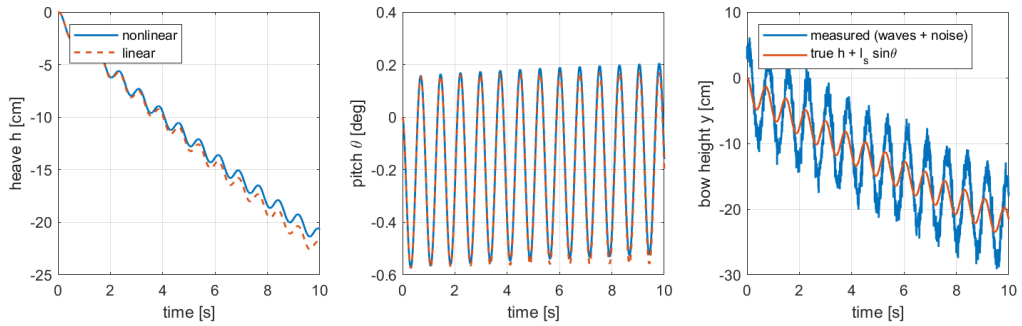


Figure 5: Regular head waves, no control: heave, pitch, and the measured bow height compared with the true value $h + l_s \sin\theta$.

11.6 Complete Simulink model

Figure 6 runs the Simulink model with all effects enabled: actuator limits, waves and sensor noise, and a 1° downward pulse on the front foils lasting 0.2 s. The applied angle lags the command because of the servo lag and the rate limit. The linear and nonlinear heave stay within a few millimetres of each other over the three seconds shown.

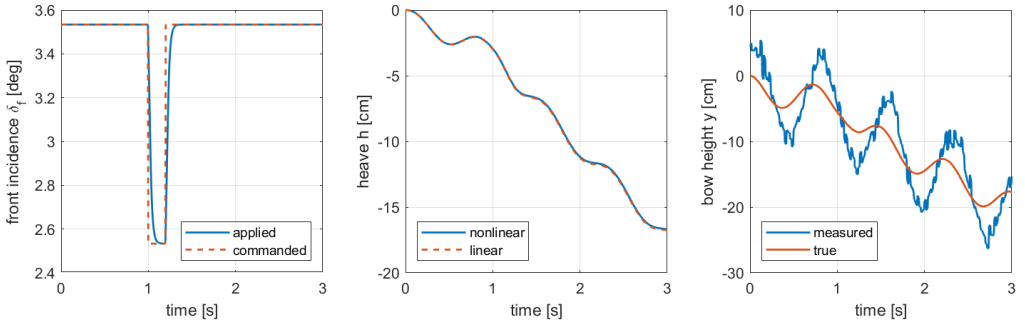


Figure 6: Simulink model with actuator limits, waves and sensor noise: commanded and applied front-foil angle, heave, and measured bow height.

12 Limitations of the model

The LTI model is valid for small deviations about level flight at one speed. It does not represent take-off and landing, surge dynamics, unsteady (Theodorsen) lift, downwash of the front foils on the rear foil, foil ventilation, or the horizontal wave orbital velocity, which modulates the dynamic pressure by about 3 % in the test sea. Wave velocities are evaluated at the trim depth and the

trim foil positions. The nonlinear model of Section 4, with saturation, rate limits, stall and sensor noise, is used in simulation (Section 11) and will be used to test the controller outside the linear range.

A Derivation of the free-surface factor

At high Froude number gravity has no time to act on the free surface, and the linearised free-surface condition reduces to zero velocity potential on the undisturbed surface. This condition is met by placing an image of the foil at the same distance d above the surface, with circulation of the *same* sense (a rigid wall would require the opposite sense). The foil and its image form a biplane with gap $2d$.

Represent the foil by a single bound vortex of circulation Γ at the quarter-chord point and enforce flow tangency at the three-quarter-chord point, a distance $c/2$ downstream. The downwash there is the sum of the contributions of the foil vortex and of the image vortex, which lies $c/2$ upstream and $2d$ above:

$$\frac{\Gamma}{2\pi\,(c/2)} \;+\; \frac{\Gamma}{2\pi}\,\frac{c/2}{(c/2)^2+(2d)^2} \;=\; U\alpha .$$

Solving for Γ and dividing by the deep-water value $\Gamma_\infty = \pi c U \alpha$ gives

$$K = \frac{\Gamma}{\Gamma_\infty} = \frac{1}{1+\dfrac{c^2}{c^2+16d^2}} = \frac{1+16\,(d/c)^2}{2+16\,(d/c)^2},$$

which is (6). Since lift is proportional to circulation, the same factor multiplies the lift. The result assumes a two-dimensional thin foil, small angle of attack and infinite Froude number; it does not model ventilation.

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[2] J. Ren and Y. Yang, "Fuzzy gain scheduling attitude control for hydrofoil catamaran," in *Proc. American Control Conference*, Boston, MA, 2004, pp. 1103–1108.

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